Milestone 1 - Data Description

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Abstract

This project analyzes the CDC Diabetes Health Indicators dataset derived from the BRFSS to examine associations between lifestyle, access to care, and diabetes. The dataset contains 23 columns including one identifier, 15 binary indicators, four ordered categorical fields, and three numerical measures. The cleaning process was minimal because there are no missing values in the dataset to begin with,

1. Data Description

The CDC Diabetes Health Indicators dataset compiles Behavioral Risk Factor Surveillance System (BRFSS) responses from U.S. adults into a large, structured table designed for risk modeling. It contains about a quarter million records with 20+ health-related variables that are mostly binary or ordinal, covering cardiometabolic conditions (such as high blood pressure and high cholesterol), health behaviors (such as smoking and physical activity), self-reported status (including general, mental, and physical health), functional limits (such as difficulty walking), and basic demographics such as age group, sex, education, and income. Body mass index shows up as a numeric field, while most of the other fields use 0/1 or small integer codes that reflect the survey answers. The label indicates whether a person has diabetes based on the answers of their survey, and the other columns combine medical risks (like blood pressure and BMI) with care access and lifestyle factors (like checkups, smoking, and exercise) to reflect everyday influences on diabetes. In general, the dataset presents a glimpse of population health indicators that are reliable for classification and exploratory epidemiologic work.

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2. Variable Descriptions

The dataset has 23 variables total: 1 identifier (ID), 15 binary indicators (Diabetes_binary, HighBP, HighChol, CholCheck, Smoker, Stroke, HeartDiseaseorAttack, PhysActivity, Fruits, Veggies, HvyAlcoholConsump, AnyHealthcare, NoDocbcCost, DiffWalk, Sex), 4 ordered categorical (“factor”) fields (GenHlth 1–5, Age 13 buckets from 18–24 to 80+, Education 1–6 from no school to college 4+ years, Income 1–8 from <\$10k to ≥\$75k), and 3 numerical counts/continuous fields (BMI, MentHlth, PhysHlth). For modeling, the working feature set is 22 columns (everything except ID): 15 binary, 4 ordered categorical, and 3 numeric. For context, Diabetes_binary shows whether the respondent reports diabetes (prediabetes/diabetes coded as 1), serving as the prediction target. ID is just a unique row key. HighBP and HighChol show lifetime diagnoses of high blood pressure and high cholesterol, while CholCheck records whether cholesterol was checked within the past five years (basic cardiometabolic screening behavior). BMI is body mass index from height/weight and acts as a proxy for adiposity. Smoker represents ever smoking at least 100 cigarettes (standard BRFSS “ever smoker”), a risk factor for vascular disease that tracks with diabetes risk. Stroke and HeartDiseaseorAttack (CHD or MI) represents major cardiovascular history that co-travels with metabolic disease. PhysActivity shows any leisure time physical activity in the last 30 days, and the diet markers Fruits and Veggies represent whether fruit and vegetables are consumed at least once per day; simple indicators for healthier eating. HvyAlcoholConsump records heavy drinking (men >14, women >7 drinks/week), which is relevant to metabolic and liver health. AnyHealthcare shows whether the person has any health coverage, while NoDocbcCost records whether cost prevented a needed doctor visit in the past year (both provide cognizance of access and potential delays in diagnosis or management). GenHlth is a five-level self-rating of overall health (excellent→poor). MentHlth and PhysHlth count days (0–30) in the past month when mental or physical health was “not good,” which often correlate with functional status and chronic disease burden. DiffWalk records serious difficulty walking or climbing stairs, a limitation linked to frailty and advanced disease. Demographics—Sex (0=female, 1=male), Age (13 ordered groups), Education (1–6 ladder of attainment), and Income (1–8 income brackets) all

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provide socioeconomic and life-stage context; these are all useful but also sensitive attributes that should be handled carefully in analysis and reporting.

3. Challenges for Analysis

The first step in preparing the dataset for analysis was to check for any null or missing values. Using `df.isnull().sum()`, we confirmed that there were no missing entries in any of the columns. All the columns returned 0 as the sum for null values. There was a possibility that the data could have had empty strings and single spaces but there were none in this data set. The lambda function to check for the empty strings and the isnull() helped to ensure that this data set was not missing any values. It was also important to verify that each column contained more than one unique value, to make sure every variable contributed meaningful information. This was confirmed by examining the number of unique values in each column using `df.nunique()`, which showed that all columns had multiple unique values. To get an initial overview of the dataset, we printed the shape of the DataFrame as well as the first and last few rows using `df.head()` and `df.tail()`.

Next, we reviewed the data types of the columns to improve the dataset's usability. When the data was initially loaded without pandas, many columns were either binary or integer types. However, after converting the dataset into a data frame using pandas, all columns were seen as integers by pandas. To make the dataset easier to understand, we converted the columns representing binary features from numeric codes (0 and 1) to more descriptive string values. Specifically, binary columns such as `HighBP`, `HighChol`, `CholCheck`, `Smoker`, `Stroke`, `HeartDiseaseorAttack`, `PhysActivity`, `Fruits`, `Veggies`, `HvyAlcoholConsump`, `AnyHealthcare`, `NoDocbcCost`, and `DiffWalk` were mapped so that 0 became `'No'` and 1 became `'Yes'`. This change helped make the data more intuitive to interpret.

Additionally, some categorical variables that used numeric codes were updated with descriptive labels to improve readability. For example, the `Sex` column was mapped from 0 and 1 to the strings `'Female'` and `'Male'`, respectively, making the demographic data clearer. The `GenHlth` column, which rates general health on a scale from 1 to 5, was converted into categories such as `'Excellent'`, `'Very Good'`, `'Good'`, `'Fair'`, and `'Poor'`. The `Age` variable was also mapped from numeric codes to meaningful age ranges: 1 to `'18-24'`, 2 to `'25-29'`, 3 to `'30-34'`, 4 to `'35-39'`, 5 to `'40-44'`, 6 to `'45-49'`, 7 to `'50-54'`, 8 to `'55-59'`, 9 to `'60-64'`, 10 to `'65-69'`, 11 to `'70-74'`, 12 to `'75-79'`, and 13 to `'80+'`. This mapping provided a clear and easy-to-understand representation of age groups. Finally, the education and income columns were converted from numeric codes to meaningful categories that reflect education levels (from no schooling to college graduate) and income brackets (from under $10,000 to $75,000 or more). These mappings improved the overall readability of the dataset, making it easier to analyze and interpret.

References

Langley, P. Crafting papers on machine learning. In Langley, P. (ed.), *Proceedings of the 17th International Conference on Machine Learning (ICML 2000)*, pp. 1207–1216, Stanford, CA, 2000. Morgan Kaufmann.

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